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Circumcision, autonomy, and public health

Introduction

Male circumcision refers to the partial or total removal of the penile prepuce (foreskin) (Cold and Taylor, 1999; Taylor et al., 1996). There is little disagreement among ethicists about the legitimacy of therapeutic circumcision—i.e., circumcision intended to address an extant pathology that has not responded to conservative treatment—nor is there much debate over the ethics of adult circumcision insofar as consent is fully informed (a condition that is not always met; Fox and Thomson, 2010; Sidler et al., 2017). What does engender controversy are circumcisions performed on healthy babies or young boys who are not old enough to provide their own consent, most commonly for sociocultural or religions reasons rather than out of medical necessity (Benatar and Benatar, 2003; Ben-Yami, 2013; Darby, 2015a; Davis, 2001, 2013; DeLaet, 2009; Earp, 2013; Foddy, 2013; Fox and Thomson, 2006; Jacobs and Arora, 2015; Mazor, 2013; Savulescu, 2013; Svoboda et al., 2016; Ungar-Sargon, 2015; Van Howe, 2013). In a recent paper, Azim McMath (2015a, p. 687) argues that while much of the controversy surrounding such non-therapeutic circumcision stems from disputes about empirical matters, some of it has to do with more theoretical questions concerning the autonomy of the child:

> while still an infant, the child cannot act autonomously. Nor has he ever been able to do so. However, if all goes well, he will eventually develop the capacity for autonomy. In addition, many of the professed benefits of circumcision, such as reduced risk of HIV infection, will occur primarily after the child becomes autonomous. Furthermore, there may be alternative means of achieving similar benefits, such as the responsible use of condoms, which depend largely on the child's future autonomous choices. In these circumstances, what does respect for autonomy require?

McMath argues that while a child's interest in future autonomy should generally be respected in relation to his own interests, the well-being of other parties may require that his autonomy be overridden in the interests of public health. At the same time, McMath seems conflicted about whether the seriousness of the threat of HIV, especially in developed countries, can in fact justify the sacrifice of individual freedom that is entailed by circumcision in infancy or

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early childhood (that is, the freedom to make one's own decision about whether to undergo an elective genital surgery at an age of understanding) (McMath, 2015b). In this context, McMath's discussion about the child's interest in making decisions that reflect his mature preferences and values when he is older is compelling (McMath, 2015a) (p. 689). But when considering arguments for paternalism in the name of public health, we suggest that McMath moves too quickly from certain empirical premises to associated policy proposals, skipping over gaps in evidence as well as important questions of value.

We do not mean to pick on McMath. In fact, his paper represents a good-faith attempt to navigate between the extreme and often polarized positions that have been articulated in the recent literature (for discussions, see Collier, 2012; Earp, 2015a). Even so, we claim, he adopts some of the same problematic assumptions about both the available data concerning health benefits and risks associated with circumcision, as well as their ethical implications, that have come to characterize the wider debate (Fox and Thomson, 2010). Using McMath's paper as a case study, then, our present aim is to slow the reel down in order to examine some of the intervening steps between data analysis and practical healthcare recommendations that must be taken into account.

Such an exercise, we hope, will have relevance not only for the debate over circumcision but also for other areas of health policy where there may be trade-offs between attempts at health promotion and respect for individuals' autonomy. In the course of our argument, we will touch on some of the main ongoing ethical disagreements concerning the permissibility of performing non-therapeutic infant or childhood male circumcisions, particularly in light of conceptual questions about the nature of a child's right to bodily integrity. First, however we provide some historical background in order to frame the contemporary debates.

Background

Male circumcision originated in prehistory as a ritual practice (Glick, 2005). It was long regarded by its supporters as conferring primarily symbolic, spiritual, or social benefits. During the Victorian period, however, doctors in Britain and the United States introduced circumcision not only as a measure of moral hygiene—dulling the sexual organ was thought to discourage childhood masturbation—but also as an intended preventive against certain physical diseases in adulthood (Darby, 2005). Routine circumcision reached a peak in

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popularity in the mid-twentieth century, but was subsequently rejected by the majority of child health authorities by the 1970s. Since that time, with the exception of the United States, it has largely been abandoned in the Anglophone countries that originally took it up (Gollaher, 2000; Wallerstein, 1985).

In the last few decades, facing increased scrutiny from legal theorists and medical ethicists as well as criticism from human rights advocates (Abu-Sahlieh, 1994; Boyle et al., 2000; Cruz et al., 2003; Denniston and Milos, 1997; Price, 1997; Sommerville, 2004), supporters of circumcision have sought to revive the 'medical benefits' narrative, casting the procedure as a secularly-defensible measure of individual and public health, as opposed to solely a religious practice (Glick, 2005; Gollaher, 2000; Miller, 2002). The search for benefits has been most fruitful in the United States, where, as noted, circumcision was adopted as a quasi-medicalized cultural norm in the late nineteenth century (Gollaher, 1994; Hodges, 1997). The acceptance and perpetuation of the procedure by key actors within the U.S. medical community at that time and thereafter established the procedure as a ubiquitous birth custom, viewed as no more remarkable than clipping the umbilical cord after delivery. Consequently, the circumcised penis became the social norm (Hodges, 1997; Waldeck, 2003). There is now a vast literature, disproportionately generated by American doctors and researchers, purporting to show that circumcision is at least partially protective against a wide range of diseases and other problems. These range from urinary tract infections (UTIs) in early childhood, through sexually transmitted infections—including HIV—after sexual debut, to penile cancer in old age (AAP, 2012; Hodges, 1997; Wallerstein, 1980).

Against this backdrop, a prominent feature of much current circumcision advocacy has been the underlying assumption that the biomedical evidence concerning such health benefits 'speaks for itself.' Once the data are available, that is, many advocates seem to assume that certain medical or public health policies should follow automatically as a logical consequence (Schoen, 2006; Tobian and Gray, 2011; for discussion, see Earp and Darby, 2015). This tendency has been particularly apparent in arguments that rely on evidence from three randomized controlled trials (RCTs) carried out in the early 2000s in Sub-Saharan Africa (Auvert et al., 2005; Bailey et al., 2007; Gray et al., 2007). These RCTs provided evidence that voluntary circumcision of adult men may reduce the risk of female-to-male HIV transmission in areas with high rates of such heterosexual transmission and a low baseline prevalence of circumcision. As Aaron T. Norton argues (Norton, 2017, p. 3), the

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epidemiologists who conducted these trials, upon seeing the results, also saw a direct path to policy recommendations, their arguments reflecting "an assemblage of scientific and ethical concerns."

Stressing that circumcision had the potential to save lives, for example, the epidemiologists urged that to refuse to act was to act unethically. According to Nathan Hodson and Susan Bewley (2017), such alarmist rhetoric has featured prominently in many policy proposals surrounding HIV/AIDS, leading to a climate of 'AIDS exceptionalism' (i.e., the tendency to coopt the language of ethics to promote urgent action—and suppress dissent—without fully considering the moral dimensions of the problem or alternative ethical or policy views). Moreover, the epidemiologists seemed to believe that

> science is the arbiter of ethical concerns via projections about future infections that rely not only upon modeling studies that employ the RCTs' statistical results, but also evidence of the foreskin's biomolecular susceptibility. If we know that foreskin is susceptible to infection by nature and that cutting it off saves lives, the implication is that not doing so kills, foregrounding the cause of death not in sexual behavior *per se*, nor in broader structural conditions, but in the body's own biological properties. (Norton, 2017, p. 3)

Faced with the RCT evidence, many public health officials, medical practitioners, journalists and even some bioethicists joined the rush to engage in extrapolation (Fox and Thomson, 2010). For example, if circumcision can lower the risk of so serious and, in some contexts, so prevalent a disease as HIV/AIDS, then, some seemed to think, it must be both medically desirable and ethically acceptable to perform it anywhere (Bossio et al., 2014, 2015; Fox and Thomson, 2010). Moreover, if it 'works' for consenting adults, it should also be performed on infants and young boys (see Bossio et al., 2015); and not only in response to individual (parental) demand, but through high-pressure public programs (Katisi and Daniel, 2015). In an interview (Norton, 2017, p. 3), one epidemiologist stated: "A lot of men are calling it a mutilation. It's like, it *is*, but it saves lives, you know? There's a basis for it." In this way, Norton remarks, "concerns about the foreskin's loss pale in comparison to circumcision's ability to prevent death."

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In light of such thinking, mass circumcision as a strategy for HIV/AIDS control is now in full swing throughout the African continent, with the generous backing of American funders such as the Bill & Melinda Gates foundation (Bell, 2015; Garenne et al., 2013; Giami et al., 2015; Katisi and Daniel, 2015). Moreover, it has been construed as a straightforward instance of applying the discoveries of modern science to human betterment, akin to vaccination campaigns for polio or clean water for the prevention of cholera (Bill & Melinda Gates Foundation, 2017; Coghlan, 2009). But the relationship between evidence of health benefits associated with an intervention in a particular context, and the appropriate health policy to pursue on the basis of that evidence, whether in the original context or elsewhere, is not straightforward (Cartwright, 2007, 2011). As the philosophers of science Nancy Cartwright and Kathryn Joyce point out (Cartwright and Joyce, 2017, p. 3):

> studies don't produce policies. Rather, studies test hypotheses about whether an intervention will produce an effect [under certain conditions]. If the intervention yields a positive result—it produces the effect—in several well-conducted RCTs, we can say that the studies support the claim that an intervention works in the study settings. Now, we have a fact. But when will this fact count as evidence supporting use of an intervention?

Answering her own question in a separate paper (Cartwright, 2011, p. 1401), Cartwright writes that "for policy and practice we do not need to know 'it works somewhere.' We need evidence for 'it-will-work-for-us' claims: the treatment will produce the desired outcome in our situation as implemented here." For this, in turn, we need "a large and varied inductive base … lots of RCTs from different populations—plus reason to believe the observations are projectable, plus an account of the range across which they project." In other words, while *efficacy* might be demonstrated under ideal/research conditions, there is no guarantee that an intervention will be *effective* in a real world setting. Thus effectiveness trials, also known as translational research, are typically undertaken before generalized policy positions are taken. No such translational research was performed in regards to male circumcision and HIV infection.

Some advocates of circumcision seem to appreciate the need for a basis for projection. That is, they recognize that even if the RCT findings are taken at face value, they are not sufficient to show that circumcision will be effective—much less prudent on public health or ethical

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grounds—in other settings apart from Sub-Saharan Africa. This realization has led to "attempts to generalize the results of the circumcision RCTs to novel contexts via appeals to the foreskin's biomolecular susceptibility" (Norton, 2017, p. 4). In other words, circumcision advocates have begun to argue that simply having a foreskin should be "effectively considered a pre-disease state defined in relation to HIV risk" (Norton, 2017, p. 5).

This conceptual shift represents a radical departure from how risk is normally understood. Unlike the identification of pre-cancerous cells in the cervix, say, that may be associated with an increase in the future risk of cervical cancer, "arguments that the foreskin poses a risk do not rely on claims that the foreskin's cells have *become* compromised, but rather, on claims that the foreskin's 'normal' cellular state is *already* risky" (Norton, 2017, pp. 5-6). Construing a healthy body part as in-and-of-itself a source of risk has proven strategically valuable for circumcision advocates: "if HIV susceptibility is an inherent feature of the foreskin, it becomes easier to dismiss contextual factors that may have shaped the RCT results, and thus generalize not only from trial context to the real world, but across real-world contexts" (*ibid.*)

It is true that human genital mucosa, whether found on the bodies of males or females, may be penetrated by HIV (Cold and Taylor, 1999; Hladik and Hope, 2009; Taube et al., 2007). It is not implausible, therefore, that reducing the surface area of such mucosa—in males, by excising the foreskin—would lower a person's risk of becoming infected with HIV by some amount, just as excising any other body part would reduce the risk of some disease affecting it, or introduced to other body sites through it (Earp et al., in press; Kluge, 1994). But to slide from such a biological assertion to the proposition that circumcision is therefore on balance desirable as a personal or public health intervention is a non sequitur (Darby, 2016). There are many possible responses to such claims, and the appropriate response will be determined not by science by in and itself, but also by the relevant values of the individuals and societies in question (Douglas, 2009).

The very notion of what constitutes optimal health, for example, is subject to cultural variation (Levesque and Li, 2014), and a preference for (say) a modest reduction in the absolute risk of a treatable infection over certain kinds of sexual sensation or experiences of bodily integrity is also a culturally influenced priority (Dekkers et al., 2005; Dekkers, 2009; Earp, 2017a, 2017b; Earp and Darby, 2017). It is telling that in countries where circumcision

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is the norm, such as Saudi Arabia or the United States, health professionals tend to commend circumcision as a prophylactic against HIV (AAP, 2012; Alkhenizan and Elabd, 2009; see also Earp, 2015b); whereas in those countries and regions where surgically unmodified male and female genitalia are the norm—such as most of Europe, Britain, Canada, Brazil, Australia, and New Zealand—health professionals do not regard circumcision as favourably and have by and large rejected it as an HIV control tactic (BMA, 2004; CPS, 2015; Earp and Shaw, in press; Frisch et al., 2013; KNMG, 2010; RACP, 2010).

Public health crises, such as the syphilis scare in nineteenth century Britain no less than HIV/AIDS in contemporary Africa, have historically given rise to demands for drastic action in which the autonomy interests of individuals become subordinated to the greater good, with an attendant rise in the level of state or social paternalism (Darby, 2015b; Gonsalves and Staley, 2014; Hodges, 1997). There are certainly situations in which the demands of public health will require the restriction of individual autonomy—for example, quarantine in the case of highly infectious diseases; and even then governments must be on guard against overreaction (Cetron and Landwirth, 2005; Drazen et al., 2014)—but it does not necessarily follow that the HIV/AIDS crisis in Sub-Saharan Africa is an instance of this proposition, nor that it warrants the heavy-handed promotion (Katisi and Daniel, 2015; Rudrum et al., 2017) of an irreversible genital alteration whose risks have not adequately been studied (AAP, 2012; Earp, 2015c; Earp and Shaw, in press; Frisch and Simonsen, 2015a, 2015b, 2016). Nevertheless, respect for autonomy entails that adult men should be allowed to elect circumcision for themselves, whether as an attempted form of prophylaxis against HIV or for any other reason. Problematically, however, as adult male volunteers have failed to materialize in sufficient numbers to meet the quotas set by circumcision backers, advocates are now pressing for the procedure to be performed on infants and young children, who cannot decline (Bailey et al., 2017; Hatzold et al., 2014; Mavhu et al., 2012; Sidler et al., 2017).

Some researchers have begun to resist what they perceive as an ill-advised circumcision policy juggernaut. For example, Kenneth Rochel de Camargo and colleagues have recently argued that the current approach to policy is both technocratic and authoritarian, running roughshod over the complexities of culturally sensitive and ethically responsible medical decision-making (de Camargo et al., 2013; de Camargo et al., 2015; de Camargo et al., 2015b; Giami et al., 2015). Inadequate attention is given to social, behavioural, and other

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contextual factors that are relevant to disease prevention and management (Darby, 2015b; Giami and Perrey, 2012; Norton, 2013, 2017; Van Howe and Storms, 2011). Bioethical and human rights principles are dismissed as nebulous (Fox and Thomson, 2010). And 'subjective' issues such as individual preferences and values regarding bodily aesthetics and genital integrity are deemed to be irrelevant (Darby and Cox, 2009; Earp and Darby, 2017; Richters, 2009).[1]

Circumcision is not, however, 'just another' public health intervention, based primarily on evidence of net benefit rather than harm. As Peter Aggleton observes, circumcision "has its roots in the deep structure of society. Far from being a simple technical act, even when performed in medical settings, it is a practice which carries with it a whole host of social meanings" (Aggleton, 2007, p. 15). These meanings may relate to rites of passage, religious customs or beliefs about hygiene, and are often "a potent indicator of hierarchy and social difference." Circumcision, then, is not a "value neutral" act, but "nearly always a strongly political act, enacted upon others by those with power." It is for this reason that close ethical scrutiny of proposals by "those with power" to enforce circumcision on the bodies of those without it is required.

A right to bodily integrity?

In order to set the stage for a potential trade-off between child autonomy and public health, McMath raises the issue of a 'right to bodily integrity.' Critics of non-therapeutic circumcision sometimes claim that the procedure violates a child's right to bodily integrity (Androus, 2004; Denniston et al., 2007; Ungar-Sargon, 2015), while supporters deny this (Benatar and Benatar, 2003; Mazor, 2013). Resolving this dispute is important because if non-therapeutic circumcision does violate a child's rights—whether to bodily integrity or anything else—its use as a public health measure would be much more difficult to justify.

There is disagreement in the bioethical literature about whether children have an absolute right to—as opposed to, say, a very strong interest in—having their bodily integrity preserved

[1] Another problem with the current approach is that it ignores ongoing concerns about the reproducibility of basic findings across the biomedical and social sciences (Earp and Trafimow, 2015; Begley and Ioannidis, 2015; Collins and Tabak, 2014; Ioannidis, 2005, 2017), often taking contentious claims for granted while ignoring or downplaying concerns about researcher bias, politicking in peer review, and agenda-driven science in this and other areas (Bero, 2017; Earp, 2016a, 2016b; Earp and Darby, 2015; Easterbrook et al., 1991; Ploug and Holm, 2015; Shaw, 2014; Smith, 2010; Van Howe, 1999, 2017).

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(Ludbrook, 1995; Mazor, 2013; Ungar-Sargon, 2015). But even if one simply grants that children do have such a right, it would not entail that one must never break the skin or otherwise alter the body of a child. As can be seen with so-called 'medically necessary' surgeries (such as might be needed to correct a heart defect, or even a therapeutic circumcision in the event of severe frostbite or incurable balanitis xerotica obliterans), it is usually permissible to infringe upon the bodily integrity of the child if doing so is clearly in the child's best interests (Kopelman, 2007; Maslen et al., 2014).

What counts as being in the child's best interests, however, is a matter of dispute (Earp and Shaw, in press; Kopelman, 2014; Kopelman and Kopelman, 2007). McMath raises the example of vaccination. As he notes, this an intervention that breaks the child's skin without being medically necessary, and yet is widely considered to be consistent with the child's best interests (and certainly not a rights violation). Insofar as this observation is meant to pave the way for the acceptability of circumcision, however, the analogy is strained (Lyons, 2013). Those who argue for a 'right to bodily integrity' do not typically hold so sensitive a notion of bodily integrity in mind that they would regard it as being impermissibly infringed by a mere pinprick on the arm (especially when the prick is needed to deliver medicines that are expected to boost the child's immunity to disease while she or he is still in childhood). Rather, they have in mind a more robust sense of bodily integrity: the absence of substantial changes to the body that may reasonably be regarded as a harm (Earp and Darby, 2017; Jonsen, 1978).

Parents who alter their children's bodies—as with nontherapeutic genital surgeries—typically do not intend to harm them. Rather, in most cases, they view the alteration as an enhancement (i.e., something that is expected to improve the child in some way, whether physically, spiritually, or socially; Earp, 2016c; Svoboda, 2013). But many children—female, male, and intersex—whose genitals were altered without a strict medical need, grow up to regard the intervention or the associated bodily change as a diminishment or even a mutilation (Bossio and Pukall, 2017; Earp and Steinfeld, 2017; Hammond, 1999; Hammond and Carmack, 2017; Lightfoot-Klein et al., 2000). This phenomenon has inspired a worldwide 'genital autonomy' movement that is largely fueled by such resentful individuals (DeLaet, 2012; Silverman, 2004; Wisdom, 2012). Such a polar 'flip' from intended enhancement to experienced net harm or mutilation appears to occur in a minority of cases across sexes (albeit with considerable variance depending on the context), especially when the sociocultural norms upholding such practices are relatively widespread and not typically questioned (Abdulcadir

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et al., 2012; Dreger, 1999; Earp, 2017a; Earp et al., in press; Gage and Van Rossem, 2006; Karkazis, 2008; Reis, 2009; Shweder, 2013, 2016; Sircili et al., 2010). Nevertheless, such extreme negative feelings appear to be more common as a response to non-therapeutic early childhood genital modifications than to other commonly-cited forms of intended pediatric enhancement—such as cosmetic orthodontia—that also involve making permanent physical changes (Bossio and Pukall, 2017; Earp, 2015d, 2015e; Hammond and Carmack, 2017; Mollov et al., 2010). Why might this be so?

One possible explanation is that alterations that by necessity damage or remove sensitive genital tissue—or more generally, tissue that it is reasonable to regard as having value in and of itself (Kipnis and Diamond, 1998)—are more likely later to be perceived as harmful compared to other bodily alterations or intended enhancements that do not have such a necessary effect. In simplest terms, if the guaranteed or intended outcome of the procedure is to excise tissue that one might rationally wish to have retained (see Earp and Darby, 2017), then the procedure is at a greater risk of being seen as 'mutilating' than one whose threat to such tissue is relatively minor or unintended. Thus, while cosmetic orthodontics do certainly alter the body for 'non-medical' reasons—and while there are non-trivial risks associated with such alteration (e.g., infections, bleeding, poor cosmetic outcome)—the potentially adverse physical changes are either temporary or accidental to the procedure; they are not its intended effect (Alani et al., 2015). Accordingly, there is no groundswell of resistance to aesthetic braces. More broadly, when the status of an intervention as an enhancement—as opposed to something that diminishes or causes harm—is stable across time and place and highly secure against possible changes in perspective, it is more likely to be in the child's bests interests overall (Earp, 2016c; Maslen et al., 2014).

It is the existence of this implicit, 'robust' sense of bodily integrity that is perhaps the main reason why piercing of girls' ears for cosmetic or cultural reasons does not arouse particularly heated moral arguments (Earp, 2012; Holm, 2004). Although such an intervention is non-therapeutic and does technically infringe upon the girl's bodily integrity, like cosmetic braces, it is normally considered too minimal to deserve special attention. Although there is momentary pain as well as some risk of infection, the 'final result' is a small hole that may very well close up if the child later decides she would like her earlobes to be hole-free. By contrast, male circumcision irreversibly removes between 1/3 and 1/2 of the motile skin system of the penis, consisting of highly innervated, touch-sensitive, functional

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and protective tissue (Bossio et al., 2016; Earp, 2016d; Kigozi et al., 2009; Scott, 1999; Sorrells et al., 2007; Taylor et al., 1996; Werker et al., 1998). We do not wish to defend the practice of infant ear-piercing; but in terms of degree of invasiveness and permanence, ear piercings are far removed from a surgical operation that excises a substantial amount of erogenous tissue from a psycho-sexually significant external organ. The latter, compared to the former, is much more easily classified as a 'substantial change to the body that may reasonably be regarded as a harm.'

While these considerations do not vitiate McMath's point that it is the best interests standard rather than an overriding or unqualified right to bodily integrity that is the relevant moral concern (for further discussion, see Carmack et al., 2016), they do suggest that the *burden of proof* that circumcision really is in the child's best interests is considerably higher than the burden of proof that immunization is in the child's best interests (Adler, 2016; Lyons, 2013). Among other differences, vaccination does not (a) remove erogenous or any other healthy tissue; (b) does not eliminate any bodily functions (such as the protective function of the foreskin and any and all sexual functions that involve manipulation of the foreskin); and (c) does not involve the permanent surgical modification of a body part whose altered state is not uncommonly perceived as a diminishment rather than an improvement.

Paternalism and harm to others

Having established that the focus should be on what is in the child's best interests, rather than on broad or unqualified appeals to potential rights-violations, McMath states that male children have an interest in reducing their risk of contracting HIV (McMath, 2015a, p. 687). Undoubtedly, they do. But this does not mean that they have an interest in being circumcised *in infancy* on such grounds, since there is no controlled evidence that infant circumcision—as opposed to adult circumcision—can in fact reduce transmission of HIV, much less to a meaningful degree outside of the context of Sub-Saharan Africa, where absolute rates of HIV transmission are among the highest in the world (Bossio et al., 2014, 2015; Darby and Van Howe, 2011; Frisch and Earp, in press; Sidler et al., 2008). Since one's susceptibility to sexually transmitted infections is far more strongly governed by socio-behavioral mechanisms than by strictly biological mechanisms (such as the presence or absence of a foreskin), it cannot be assumed that a partial protective effect accruing from adult

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circumcision in Sub-Saharan Africa would also follow from circumcision of infants in other contexts (RACP, 2010).

But what if it could be assumed? In other words, what if, one day, there is convincing evidence that specifically infant circumcision in some setting can reduce the risk of female-to-male transmission of HIV in that setting? Would that justify infant circumcision? No, it would not. At best it would count in favor of such circumcision—it could be added to the benefit column—but the *weight* to be assigned to this benefit would depend on many factors, both ethical and epidemiological. For example, what is the absolute risk of contracting HIV through heterosexual intercourse in the relevant context? If it is very low, as it is in most contexts outside of Sub-Saharan Africa, then the weight to be assigned in this regard should also be low. To see why this is the case, consider the following analysis of HIV transmission in the United States by Sarah Bundick (2009):

> Heterosexual contact is estimated to be responsible for only 5,250 new infections in men each year … there is no strong evidence that circumcision reduces the risk of male-to-female or male-to-male transmission via sexual contact [and it] is not related to HIV transmission by injecting drug use. Therefore, the current data suggest that the 5,250 female-to-male transmissions are the only ones [even eligible] to be prevented by male circumcision. If we assume that all 5,250 men who get HIV from a female sexual partner are not circumcised (though this is certainly not the case), the data suggest that about half of these infections — around 2,625 infections or ~5% of new infections — may have been prevented if the men had been circumcised. If we then factor in the number of men who are circumcised when they are infected (approximately 70-80% of American men are already circumcised), the number of infections that could have been prevented by circumcision drops considerably. Taken together, the data suggest that the number of HIV infections that could be prevented in the U.S. by promoting infant male circumcision is likely to be only in the hundreds per year — a tiny fraction of the estimated 50,000 new HIV infections.

Such 'tiny' numbers are based on generous assumptions about the potential effectiveness of infant circumcision as an HIV preventive in the United States because the applicability of the African RCT data pertaining to adult males is being simply stipulated. Nevertheless, they demonstrate the importance of taking context—including baseline risk figures—into

consideration when trying to assess whether (male) children have any interest at all in being circumcised as prophylaxis against HIV. But let us just assume that they do—that *some* weight should be assigned to this potential benefit of circumcision, however small. This still does not tell us whether the surgery is in the overall best interests of the child, because additional weights would have to be assigned, not only to other purported benefits, but also to risks and harms, many of which have not been accurately estimated (and some of which, such as feelings of loss or resentment, are subjective in nature and therefore impossible to quantify; see Frisch and Earp, in press).

Finally, the attitudes and values of the child himself with respect to risk-types, risk-comparisons, and alternative risk-reduction strategies also have to be factored in. For example, is a slight reduction in the absolute risk of contracting a rare and treatable infection that can also be avoided non-surgically 'worth' the risk of a similarly rare surgical error resulting in damage to the head of the penis? Is it 'worth' the risk of removing too much foreskin, leading to painful erections or penile curvature later in life? There is no objective answer to such questions. Instead, only the child himself when he is grown can assess such trade-offs, at which point he can take into consideration the fullness of his circumstances as well as his own values (Earp and Darby, 2017). What this suggests is that no amount of new evidence concerning either the benefits or risks of circumcision will be sufficient to determine the question of interests. There will always be a personal, subjective aspect to such judgments, which even McMath sees as counting in favor of refraining from elective circumcisions before an age of consent (McMath, 2015a, p. 689). He asks the reader to

> Consider the extent to which STI [sexually transmitted infection] prevention constitutes a benefit. If the child grows up to be celibate, it will not be a benefit at all; if he grows up to be sexually profligate, it may be a significant benefit. The child himself, once he reaches an age of competence, will be in a better position to judge his own interests than his parents will be while he is still an infant.

Moreover, giving the child the opportunity to decide for himself at a later age can help resolve disagreements over values. As McMath notes, reasonable people disagree over what constitutes a harm or benefit. For example, "Some people believe circumcision benefits the child by bringing him closer to God, while others disagree. In light of such disagreement,

some commentators conclude that the parents should decide" (p. 689). But this does not follow:

> After all, the child will have an interest in living according to his own values, which may not reflect those of his parents … only the child himself, when he is older, can be certain of his values. Thus, if disagreement over values constitutes a reason to let the parents decide, it constitutes an even stronger reason to postpone the decision until the child himself can decide. (McMath, 2015a, p. 689)

Given these and other considerations, McMath concludes that we should be "idealists" about the infant child's future autonomous choices insofar as they affect himself. But what if they affect others? This is where McMath brings in his argument about public health. Given the lack of any controlled evidence for a protective effect of infant circumcision against the acquisition of HIV, we can assume that McMath takes for granted something like the "biomolecular" view of foreskin risk proposed by circumcision advocates (Norton, 2017). That is, based on the unproven assumption that it is the sheer lack of a foreskin in the circumcised men that was chiefly responsible for the RCT findings—and that this lack *per se* will result in a similar risk-reduction in any epidemiological environment—McMath argues that we must consider the interests not only of the child himself, but also "anyone who, in future, has sex with him or shares needles with him" (McMath, 2015a ,p. 687). We need to do this because, according to McMath, "reducing one person's risk of contracting HIV—or any other infectious disease—also reduces the risk to many other people. Thus, we need to consider the impact of circumcision not only on the interests of the child, but also on public health."

Now, by 'public health,' McMath really appears to mean 'HIV prevention,' since he does define the term and his examples almost exclusively concern the latter. But there are several issues being obfuscated here. First, simply having been circumcised, whether in infancy or later in life, does not entail that a man's risk of contracting HIV is lower than it otherwise would be, much less that he is less likely to transmit HIV to others. Part of the reason for this is that some circumcised men engage in 'risk compensation' (Feldblum et al., 2015; Grund and Hennink, 2012; Jung et al., 2016; Kibira et al., 2017; Riess et al., 2010). This occurs when a circumcised man, presumably because he believes that his risk of contracting HIV is lower due to his lack of foreskin, comes to rely less on more surefire forms of safe sex, such

as condom use or a reduction in the number of concurrent sexual partners. If he does reduce his reliance on these measures, he may actually increase both his and his partners' risk of infection, over and above whatever advantage he may initially have had (in this respect) in being circumcised. Moreover, the only RCT looking specifically at male-to-female transmission of HIV as a function of the male's circumcision status had to be stopped early due to futility: more female partners of circumcised men, compared to intact men, were becoming infected with HIV, possibly due to early resumption of sex during wound-healing (Wawer et al., 2009). If public health or harm-to-others is the concern, then, one needs to take such considerations into account (Dushoff et al., 2011).

Failing to raise such caveats, McMath nevertheless maintains that the prospect of harm-to-others may play a legitimate role in justifying infant circumcision on public health grounds, presumably in any setting (since he does not specify an epidemiological environment). He is aware, however, that this line of thought could lead to policy conclusions that might be seen as overly paternalistic. To address this concern, he writes:

> People sometimes make autonomous decisions that result in risks and harms to others. There is nothing paternalistic about coercive intervention to prevent these other-regarding risks and harms. While people should generally be at liberty to assume risks to themselves, they should not generally be at liberty to impose risks on others. To the extent that the child's future choices may harm others, and to the extent that circumcision may help to mitigate these harms, the procedure may be justified as a matter of public health. (McMath, 2015a, p. 689)

It is true that people are not generally at liberty to impose (undue or unwarranted) risks of harm on others. But the mere possession of an intact penis does not impose risk on anybody, and certainly not undue or unwarranted risk. In other words, McMath slides invisibly in this passage from harm caused by potential "future choices" (such as failure to wear a condom), to harms caused by *simply having an unaltered penis*—i.e., the only factor that could be "mitigated" by a circumcision. But as Norton notes, "the presence or absence of a body part cannot reasonably be said to be the cause of HIV-infection, but rather plays an indirect or 'moderating' effect in the likelihood of infection that may or may not apply depending upon other conditions" (Norton, 2013, p. 276).

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What is the more general implication of McMath's suggestion? In another paper, Norton writes: "Policies aimed at making routine neonatal circumcision more accessible prioritize removing the 'vulnerable' foreskin over what kinds of sexual practices or HIV prevention strategies an infant may eventually engage in as a sexually mature person" (Norton, 2017, p. 7). In response to such observations, McMath notes that "many people do not use condoms responsibly. Consider the fact that nearly half of all pregnancies in the developed world are unintended. We cannot necessarily assume that the child will autonomously decide to use condoms responsibly when he grows up" (McMath, 2015a, p. 688).

The reference McMath gives for unintended pregnancies is to a study that made no breakdown on the basis of whether or not the male was circumcised; it may well be that unintended pregnancies are more common among the circumcised. In fact, some studies suggest that circumcised men, compared to genitally intact men, are less likely to use condoms as well as more likely to experience condom slippage when they do (Crosby and Charnigo, 2013; Frisch, 2016; Richters et al., 1995). Thus, although McMath argues that it is necessary to be a realist rather than an idealist in such matters, it is not obvious that the realist response requires us to accept the propriety of infant genital surgery, even on other-affecting grounds.

Instead, it leads us to consider how adult sexual partners might choose to negotiate the risks to which they may expose themselves: except in the case of rape, nobody is forced to have sex with a 'riskier' sexual partner.[2] Moreover, people do not have sex as embodied statistical averages: individual differences in sexual context and behavior are far more predictive of one's transmission risk than circumcision status, such that men's prospective partners would do well to focus on those other differences rather than on whether the men have had their foreskins removed (which, if they have, could create a false sense of security). To summarize, it is only consensual adult behavior—engaged in by both or all parties who may be affected—that presents a risk of harm. Sexually transmitted infections are not like sneezes: they cannot be spread throughout a school or a shopping center to unsuspecting third parties.

Tying these strands together, as Norton (2017, p. 15) notes, "promoting circumcision to reduce risk for a given infant"—much less a possible future sexual partner of his—"is based

[2] Assuming that is what the average, all-things-considered non-circumcised male is. However, it is not clear that we are justified in assuming this, especially given the higher rates of male-to-female transmission of HIV among circumcised men in the fourth RCT (Wawer et al., 2009).

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on a number of assumptions at the moment the decision is made, assumptions that reflect the emergence of circumcision status as an HIV risk category rooted not in sexual identity or behavior but in the body's own biological properties."

Yet such assumptions about risk lead to absurd consequences. Any healthy part of person's body could, in principle, be 'misused' to inflict harm on other people. For example, one's finger could be used for poking out someone's eye. But since it can also be used for morally unproblematic purposes, and it serves many important functions, infant finger-removal should not be entertained. In other words, it is problematic to remove an unoffending part of a person's body *just in case* he might use it in a way that would increase the risk of harm to others, especially insofar as those other people also possess agency in protecting themselves from any such harm.

When looking at an infant, we do not know what his "future choices" regarding sex and sexual behavior will be. If, as McMath suggests, it is nevertheless permissible to excise his healthy foreskin without his consent, on the assumption that its sheer retention on his body puts others at risk of harm, then why is it not permissible to sever an adult man's foreskin, whose future sexual behavior is much easier to predict? If the mere possession of intact genitalia is meant to be understood as 'risky' and potentially other-harming, then—on public health grounds—there is equal, if not more reason, to engage in coerced circumcision of men.

But this would be criminal assault (Merkel and Putzke, 2013). If it is not permissible to coercively circumcise a man on public health grounds, it is questionable why we should be able to circumcise an infant or small child on the same grounds, simply because the latter is pre-autonomous and can neither give nor withhold his consent. In other words, given that healthy genital tissue is *prima facie* valuable, and that the mere possession of intact body parts cannot reasonably be construed as presenting a special kind of risk, a stance of skepticism toward proposals such as McMath's seems in order.[3]

[3] It is conceivable that this analysis would differ if the foreskin had no functions or positive properties; if it were a uniquely vulnerable vector for diseases with serious public health ramifications (that could not be addressed in less ethically problematic ways); if the surgery could not be delayed without increasing the absolute risk of serious harms to an ethically decisive degree (see Earp and Darby, 2017); or if it were sound as a matter of general principle to construe healthy tissues as dangerous simply by virtue of being left on the body. Since none of these propositions hold, however, the prospect of non-consensual childhood circumcision is hard to justify from a public health perspective.

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McMath himself, writing subsequently on the *Journal of Medical Ethics Blog,* does appear to concede some of the problematic implications of his view. Responding to the release of a non-peer-reviewed draft policy by the U.S. Centers for Disease Control and Prevention (CDC) stating that the benefits of circumcision outweigh the risks (CDC, 2014), McMath (2015b) writes:

> the CDC relies far too heavily on HIV prevention as a justification for infant circumcision. In the United States, the adult prevalence of HIV is 0.6 percent – not very high by global standards. Furthermore, as the CDC itself admits, only 10 percent of new infections are transmitted sexually from a female to a male … as I have argued, most of these initial infections cannot justify infant circumcision; it is the extent to which these infections increase risks to others that is ethically relevant. All of this suggests it would take very many (perhaps thousands) of circumcisions to prevent one new ethically relevant HIV infection in the United States. The public health justification for infant circumcision, at least in terms of HIV prevention, seems much weaker than the CDC's enthusiasm would suggest.

Conclusion

McMath's basic argument—that individual rights and interests must sometimes give way to the general good—is in principle sound, but the question is whether it is correctly applied to the instance of child circumcision. Nobody but an extreme libertarian would disagree that it can be reasonable to restrict an individual's freedom of action—including with respect to certain decisions about the state of his own body—if the general welfare interests are substantial enough and the burdens placed on the individual are proportionate to the gravity of the problem and equitably distributed.

In the public health field a good example is quarantine—once a regular feature of trans-oceanic travel, and still seen sometimes today in regions of severe infectious epidemics. Under quarantine an individual's liberty is restricted for a short period in order to protect other members of society from diseases that he or she is or may be carrying. Since all individuals have an interest in freedom of movement, society or other authorities (parents in the case of children) must have good reason to override this interest. But unlike circumcision, quarantine is short-term and has no lasting consequences (provided proper access to medical

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care and other necessities); although it is a harm, it is a minor harm that imposes a short-lived burden. Once the prescribed period has elapsed, an individual can resume his life at the point where he left off; the only lasting effects are memories. It is also equitable in that all persons at risk are subject to the same restraint. Once the issue is cast in these terms it is clear that the public health justification for quarantine is not applicable to childhood male circumcision.

The inadequacy of the public health argument for circumcision of boys was well summarized in a paper as early as 2002. In "Prophylactic Interventions on Children: Balancing Human Rights with Public Health," Hodges et al. performed a systematic analysis of the rights and interests of children in relation to public health benefit (Hodges et al., 2002). With a view to laying down a set of standards on how conflicts between the demands of public health and the principles of human rights might be resolved, and observing that prophylactic interventions on children were traditionally justified on the grounds of "best interests of the child" and/or "public health," Hodges et al. proposed two sets of criteria that should be met for an intervention along these lines to be acceptable.

The criteria for the "best interests of the child" argument were (1) presence of clinically verifiable disease, deformity, or injury; (2) least invasive and most conservative treatment option; (3) net benefit to the patient and minimal negative impact on patient's health; (4) competence to consent to the procedure; (5) standard practice; (6) individual at high risk of developing the disease. The criteria for the "public health benefit" argument were: (1) substantial danger to public health; (2) condition must have serious consequences if transmitted; (3) effectiveness of the intervention; (4) degree of invasiveness of the intervention; (5) whether individual receives an appreciable benefit not dependent on speculation about future behavior; (6) the health benefit to society must outweigh the human rights cost to the individual.

The authors evaluated prophylactic mastectomy and cosmetic ear surgery against the best interests of the child criteria, immunization against the benefit to public health criteria, and childhood male circumcision against both sets of criteria. They concluded that while immunization generally satisfied the best interests and public health justifications, circumcision satisfied neither. Such an intervention was thus impermissible because it was performed on a minor without consent; the human rights cost to the individual exceeded the proven public health benefit; and the diseases from which circumcision might provide some

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level of protection could be avoided through appropriate behavioral choices or otherwise managed non-surgically.

Individuals have a substantial interest in bodily integrity (including genital integrity) and autonomy (including what medical treatments to adopt). In contrast with quarantine, the harms of circumcision (as felt by those who do regard themselves as harmed by the procedure) are permanent and irreversible—loss of an valued external bodily structure that cannot be recovered, taken without consent (Earp and Darby, 2017; Hammond and Carmack, 2017; Watson, 2014). Moreover, childhood male circumcision is inequitable in that it targets only male infants and boys (leaving females, intersex children, and adults alone), thus forcing them to bear the whole cost of whatever public health benefit is being pursued. It follows that the case for the permissibility of childhood male circumcision as a public health initiative must be far stronger than the case for the permissibility of either vaccination or quarantine—a case we have argued that McMath has not succeeded in establishing.

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